

Hyeonji Kim

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Research Interests

Keywords: Large Language Models; LLM Agents; Retrieval-Augmented Generation; Natural Language Processing; Database Query Processing; Deep Learning; Graph Databases; Big Data Processing.

- **Domain-specialized LLM agents:** multi-agent orchestration, intent decomposition and routing, and grounded tool use for high-stakes domains (insurance, finance).
- **Retrieval-augmented generation (RAG) systems:** domain-organized index design, semantic chunking, hybrid retrieval, and principled evaluation (LLM-as-a-judge, regression testing) for production QA systems.
- **LLM-driven data systems:** dataset curation, augmentation, and distillation for domain adaptation and fine-tuning; tool-augmented agents for grounded reasoning and action.
- **Natural-language interfaces to databases:** robust dataset augmentation/cleaning and principled evaluation for NL2SQL, and multi-turn dialogue systems to databases.
- **Large-scale retrieval & indexing:** hierarchical and multi-vector embeddings, cross-modal (text-image) retrieval, time-aware search, and approximate nearest neighbor (ANN) methods.
- **Event-centric intelligence:** detecting and modeling real-world events from news, social media, and enterprise data for downstream analytics and decision support.

Education

Ph.D., Convergence IT Engineering (Integrated M.S.-Ph.D.), Pohang University of Science and Technology (POSTECH), Aug 2024.

Dissertation: "Systematic Data Augmentation and Cleaning Techniques to Improve Text-to-SQL Models." Advisor: Prof. Wook-Shin Han.

B.S., Creative IT Engineering (Magna Cum Laude), Pohang University of Science and Technology (POSTECH), Feb 2016.

Appointments

AI Engineer (Manager), AI Innovation Center, Meritz Fire & Marine Insurance, Feb 2026 – Present.

Research and development of LLM-based agent systems for insurance sales and claims support.

Postdoctoral Fellow, Database & Data Mining Lab (DSLAB), POSTECH, Aug 2024 – Jan 2026.

Full-time Research Intern, Oracle Labs (PGX team), CA, Dec 2015 – Mar 2016.

Industry Experience — Meritz Fire & Marine Insurance (Feb 2026 – Present)

MeAI: AI Sales Assistant for Insurance Planners — AI Engineer, AI Innovation Center.

MeAI is an in-production AI sales assistant that answers insurance planners' (FPs') questions on coverage, claims, underwriting, sales strategy, and internal operations, grounded in company policy documents and enterprise data.

- **Agent architecture redesign (intent-graph multi-agent system).** Redesigned the query-processing pipeline from a single unidirectional planner into an intent-graph architecture: a router decomposes a multi-intent user query into a dependency graph of intents and dispatches each intent to domain-specialized expert agents (e.g., claims/coverage, underwriting, sales support), executing independent intents in parallel while propagating results along dependencies and synthesizing a single coherent answer — enabling compound questions that the previous planner could not handle.
- **Retrieval index design for domain-specialized RAG.** Reorganized retrieval corpora into domain-organized indexes with semantic-unit chunking; condensed lengthy policy documents into coverage-level summary units and connected coverages to standardized disease/procedure classification codes, substantially improving retrieval precision for grounded answers.
- **Evaluation and quality infrastructure.** Built automated quality assurance for LLM agents: pre-deployment regression gates that evaluate the exact release artifact, and nightly automated evaluation with LLM-as-a-judge scoring and day-over-day regression detection, with results persisted for monitoring. Established label-free evaluation methodology and expert-in-the-loop gold-set construction for coverage-analysis QA.

Research Projects (Selected)

Methodology for Extracting Notable Stocks from Domestic Economic News Using Generative AI — Project Lead, Apr 2025 – Jan 2026.

High-level pipeline for event detection and stock candidate surfacing (research prototype). Responsibilities: project lead, problem formulation, evaluation design, and coordination with industry partner.

SW StarLab “Development of a Conversational and Self-Tuning DBMS” — Project Lead, 2018 – Jan 2026.

Led efforts to make DBMSs usable to non-experts by building a natural-language interface to databases and automatic physical/knob tuning that simplifies performance optimization for practitioners. Responsibilities included NL2SQL modeling, dataset curation/augmentation, and evaluation infrastructure for robust QA over RDBMS.

Smart Structuring of Large-Scale Dark Data — Project Lead, 2017.

Developed a machine-learning-based framework to convert unstructured enterprise “Dark Data” into structured data. Delivered workflow components (preprocessing, candidate-set mapping, feature extraction, supervised learning and inference), a pilot on public job announcements, and an end-to-end automation plan including SQL/Python APIs.

Publications (Selected)

Kyoungmin Kim, Hyeonji Kim, George Fletcher, Wook-Shin Han. “Combining Sampling and Synopses with Worst-Case Optimal Runtime and Quality Guarantees for Graph Pattern Cardinality Estimation.” In Proc. ACM SIGMOD, 2021.

Hyeonji Kim, Byeong-Hoon So, Wook-Shin Han, Hongrae Lee. “Natural Language to SQL: Where Are We Today?” Proc. VLDB 13(10):1737-1750, 2020.

Hyeonji Kim, Juneyoung Lee, Sourav S. Bhowmick, Wook-Shin Han, JeongHoon Lee, Seongyun Ko, Moath H. A. Jarrah. “DUALSIM: Parallel Subgraph Enumeration in a Massive Graph on a Single Machine.” In Proc. ACM SIGMOD, 2016.

International Patents

Wook-Shin Han, Hyuk-Kyu Kang, Hyeonji Kim. "Method for Enhancing Learning Data Set in Natural Language Processing System." U.S. App. No. 18/725,248, June 2024.

Wook-Shin Han, Hyuk-Kyu Kang, Hyeonji Kim. "Apparatus and Method for Processing Natural Language Query about Relational Database Using Transformer Neural Network." U.S. App. No. 17/964,807, Oct 2022.

Wook-Shin Han, In-Hyuk Na, Hyeonji Kim, Hyuk-Kyu Kang. "Column and Table Prediction Method for Text-to-SQL Query Translation Based on a Neural Network." PCT/KR2019/018491, Dec 2019.

Honors & Scholarships

Google Anita Borg Scholars APAC, 2016.

POSTECH-SUNY Korea IT Consilience Creative Program (ITCCP) Scholarship, 2012–2020.

Teaching & Mentoring

Instructor, corporate lecture series on Retrieval-Augmented Generation and Agentic AI (4 sessions: IR fundamentals; RAG foundations; Advanced RAG; Agentic RAG & Evaluation), with hands-on labs, 2026.

Teaching Assistant / Instructor Support (Samsung Electronics): Data Foundation (Oct 2017), Advanced Data Expert Course (Jul–Nov 2019), Data Programming Course (Apr 2020).

Reviewing Service

ACM SIGMOD (2016, 2018, 2021–2022, 2026); VLDB (2016, 2018–2021, 2025); KDD (2017–2021); ICDE (2020–2021); EDBT (2018); IEEE TKDE (2016).

Languages & Skills

Korean (Native); English (Professional).

Programming: Python (Expert); C++ (Expert); SQL (Expert); Java (Advanced); JavaScript (Working proficiency); R (Working proficiency); CUDA C/C++ (Working proficiency).